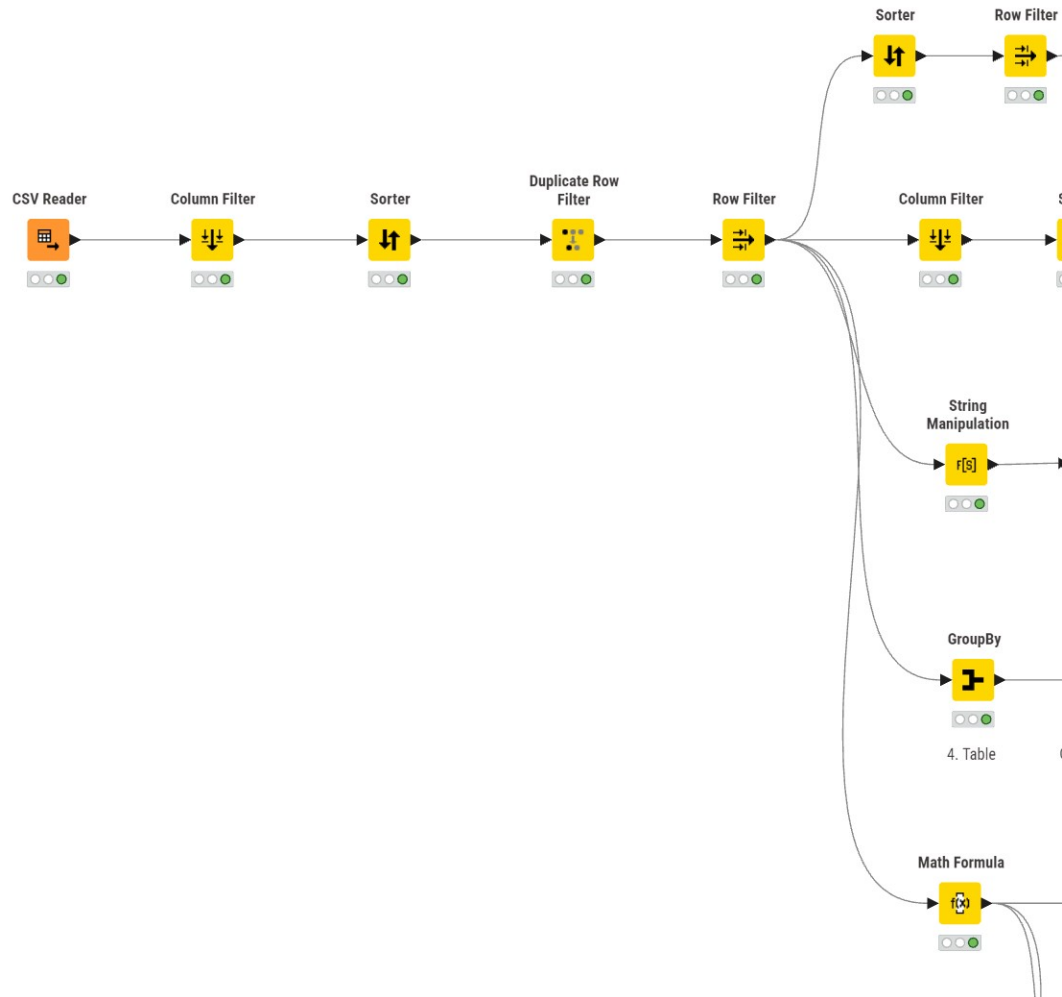


T01 (a): Build Appliance Energy Consumption website

Website: <https://cos-30045-data-visualisation-jade.vercel.app/>

T01 (b): Data Processing with KNIME



T02: Data Exploration and Basic Chart Types

1. What type of TV screen technologies are currently available in Australia and which are the most frequent?
 - a. What data is needed to answer this question?

Column Name	Data Type	Description
Screen_Tech	String	The display technology used by the TV (e.g., LCD, LCD (LED), OLED).
Availability Status	String	The market status of the

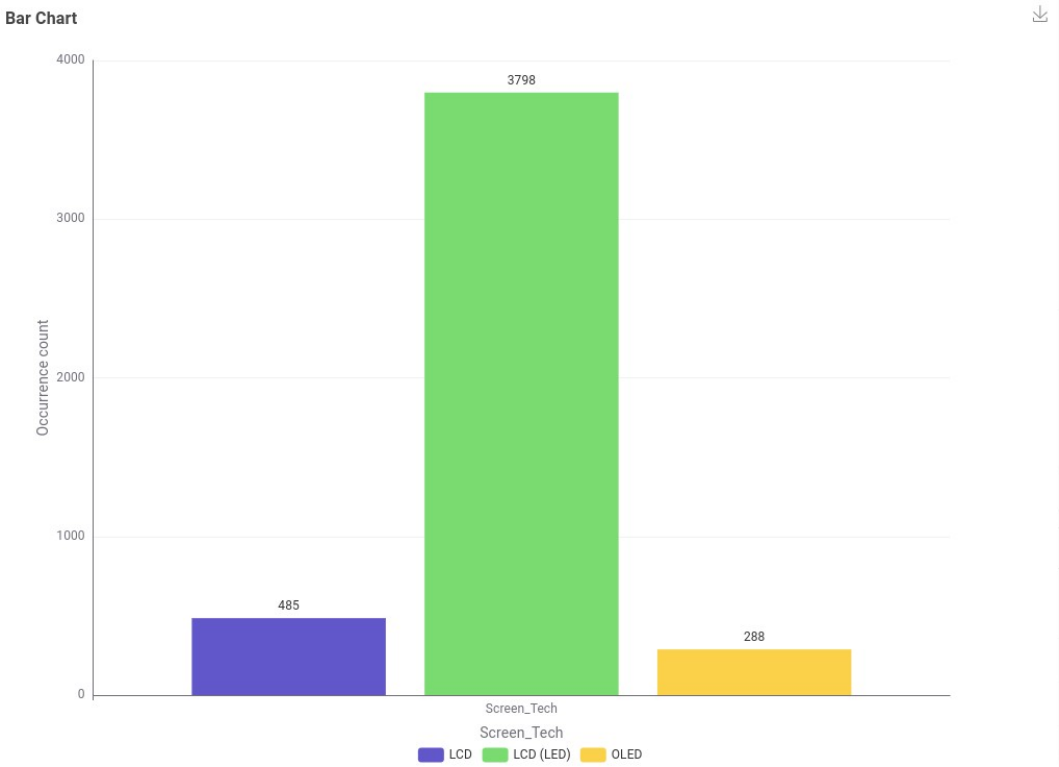
		TV (Available or Unavailable).
SoldIn	String	The countries where the TV is sold (e.g., Australia).
Submit_ID	Integer	Unique Identifier

b. What chart type would be good to represent this data and give some reasons?

Bar Chart

Reason: Bar chart offer a more distinct difference between Screen_Tech category. It is more readable compare to using pie chart.

c. Draw a sketch of the chart



d. Draw a sketch of the data table required to build the chart

Rows: 3 | Columns: 2

Sorting

Filter

Q Search

☐	#	RowID	Screen_Tech	Count*(Submit_ID)
			String	Number (Integer)
☐	1	Row0	LCD	485
☐	2	Row1	LCD (LED)	3798
☐	3	Row2	OLED	288

2. What screen sizes are currently available, and which are the most frequent?

a. What data is needed to answer this question?

Column Name	Data Type	Description
Screen size (inches)	Float	The physical diagonal measurement of the display screen, specified in inches.
Screen size (inches) -> count	Integer	The total number of records (frequency of occurrence) for that specific screen size in the dataset.

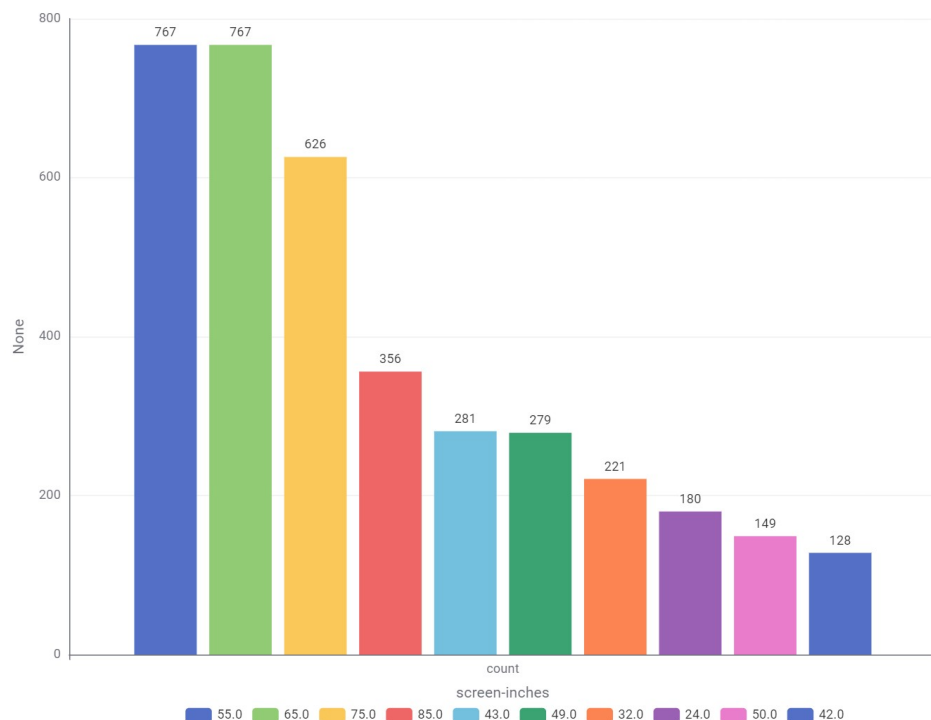
b. What chart type would be good to represent this data and give some reasons?

Bar Chart

Reason: There are many different screen size variant across the data. Using Bar Chart to view is much clearer because bar chart is the most effective choice for this data because it provides an immediate, accurate comparison of counts across multiple discrete categories.

c. Draw a sketch of the chart

Bar Chart



- d. Draw a sketch of the data table required to build the chart

Rows: 48  Sorting  Filter

☐	#	RowID	count
			 Number (Integer)
☐	1	55.0	767
☐	2	65.0	767
☐	3	75.0	626
☐	4	85.0	356
☐	5	43.0	281

Screen size currently available: 16.0, 18.0, 19.0, 21.0, 22.0, 23.0, 24.0, 27.0, 30.0, 31.0, 32.0, 34.0, 38.0, 39.0, 40.0, 41.0, 42.0, 43.0, 46.0, 48.0, 49.0, 50.0, 54.0, 55.0, 56.0, 57.0, 58.0, 60.0, 64.0, 65.0, 67.0, 69.0, 70.0, 74.0, 75.0, 76.0, 77.0, 78.0, 82.0, 83.0, 85.0, 86.0, 88.0, 97.0, 98.0, 100.0, 114.0, 116.0

Which are the most frequent: 55 and 65 inch are the most frequent

3. Which brands have the greatest number of different models?

- a. What data is needed to answer this question?

Column Name	Data Type	Description
Submit_ID	Integer	A unique numerical identifier assigned to each individual record or survey submission.
Brand_Reg (Lowered case)	String	The standardized brand name recorded in lowercase text to ensure consistent grouping and accurate count aggregation.

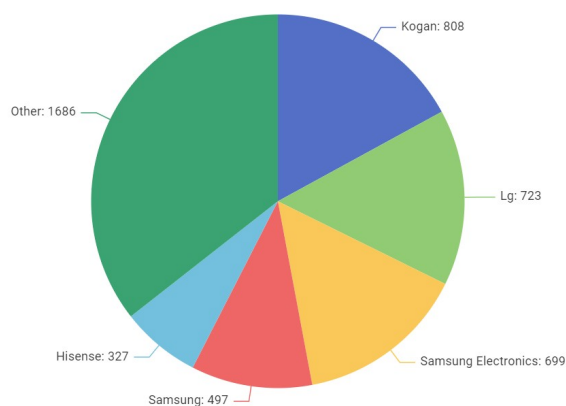
- b. What chart type would be good to represent this data and give some reasons?

Pie Chart

Reason: A pie chart clearly visualizes how each brand's share contributes to the total (100%) volume of submissions. Pie chart also effectively consolidate category because smaller brands are grouped into an "Other" segment, the chart maintains a clean, readable number of slices without visual clutter.

- c. Draw a sketch of the chart

Pie Chart



- d. Draw a sketch of the data table required to build the chart

Rows: 4740 | Columns: 2

Sorting

Filter

Q Search

☐	#	RowID	Submit_ID Number (Integer)	lowercase_Brand_Reg String
☐	2806	Row0	140130	Philips
☐	2820	Row1	140286	Philips
☐	2830	Row2	140289	Philips
☐	2852	Row3	140292	Philips
☐	2884	Row4	140296	Philips
☐	2854	Row5	140297	Philips
☐	4165	Row91	140784	Shenzhen Xinchada Technology Co., Ltd
☐	741	Row6	141068	Hisense
☐	752	Row7	141093	Hisense
☐	1202	Row20	141237	Kogan
☐	1238	Row19	141271	Kogan

Answer: Kogan with 808 (17.05%) model

4. Which type of screen technology consumes the least amount of power?

a. What data is needed to answer this question?

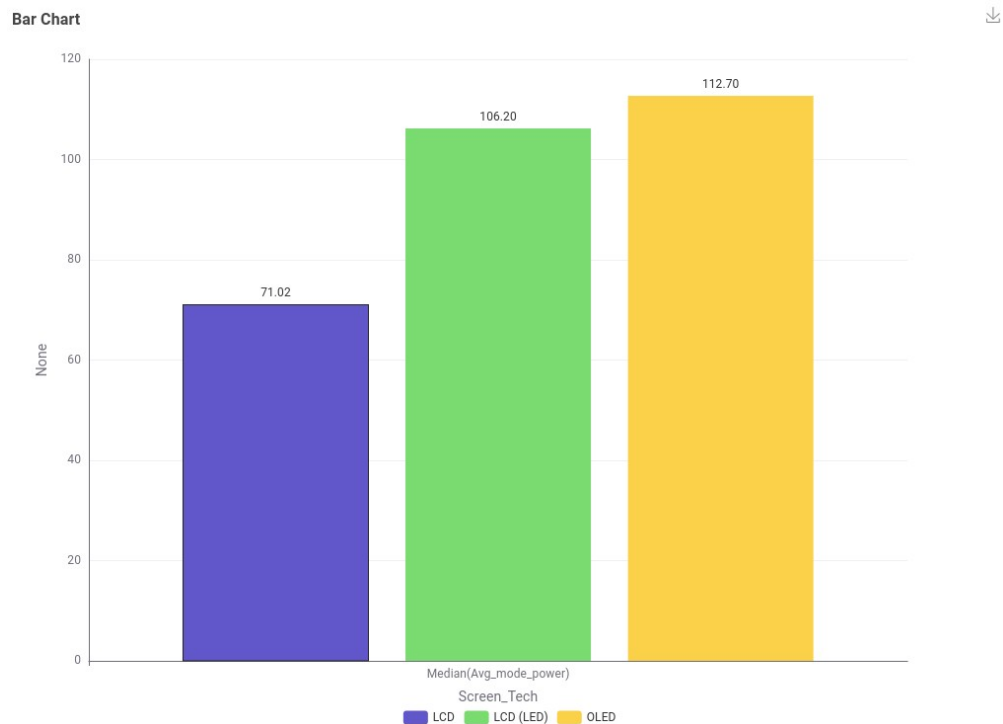
Column Name	Data Type	Description
Screen_Tech	String	The categorical display technology used by the screen (e.g. LCD, LED, OLED).
Avg_mode_power -> median	Float	The median continuous power consumption (in watts) measured across devices within each screen technology category.

b. What chart type would be good to represent this data and give some reasons?

Bar Chart

Reason: A bar chart is ideal because it compares a continuous numerical metric (median power consumption) across discrete, independent categories (screen technologies). Aligning the bars along a common baseline makes difference in magnitude immediately obvious and allows for an intuitive, direct visual ranking from lowest to highest power usage.

c. Draw a sketch of the chart



d. Draw a sketch of the data table required to build the chart

Rows: 3 | Columns: 2  Sorting  Filter

 Search 

☐	#	RowID	Screen_Tech  String	Median(Avg_mode_power)  Number (Float)
☐	1	Row0	LCD	71.024
☐	2	Row1	LCD (LED)	106.2
☐	3	Row2	OLED	112.7

Answer: LCD with the median of 71.02 mode power

5. What is the relationship between screen size and power use?

a. What data is needed to answer this question?

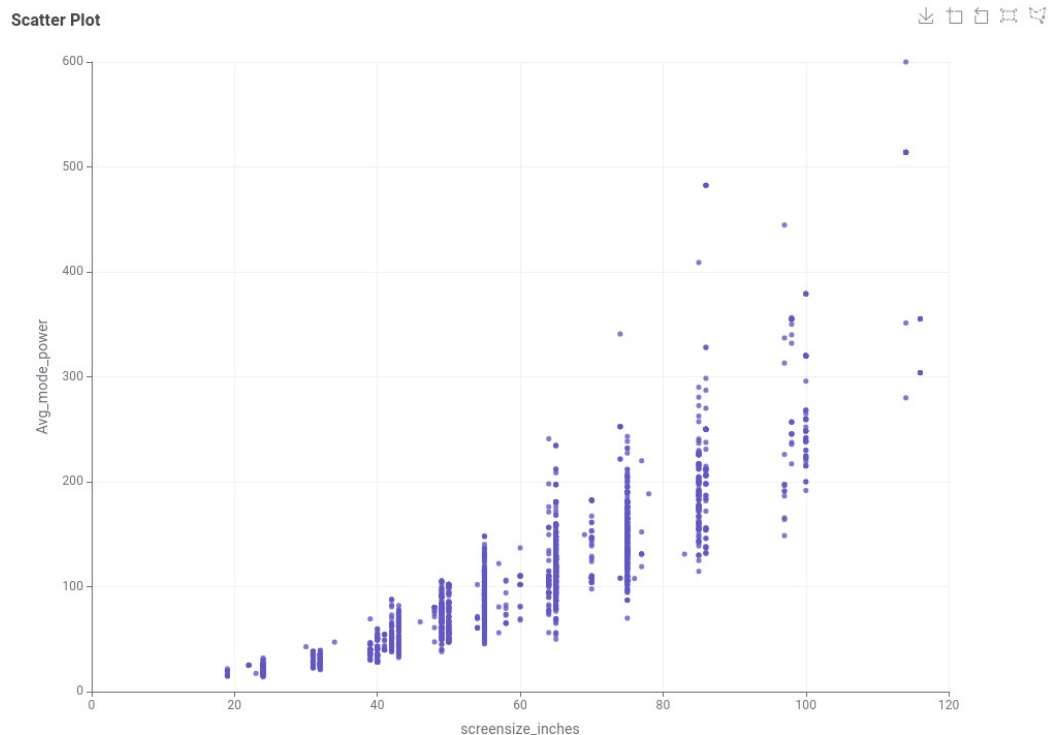
Column Name	Data Type	Description
Avg_mode_power	Float	The continuous numerical value representing average power consumption in on-mode.
screensizes -> inches	Float	The physical diagonal display dimension of each screen measured in inches.

b. What chart type would be good to represent this data and give some reasons?

Scatter Plot Chart

Reason: A scatter plot is the ideal choice because both variables are continuous numerical data, making it effective for displaying the correlation and relationship between screen size and power consumption. It clearly reveals overall trends such as power usage increasing non-linearly with screen area while preserving individual data points to highlight variance, clusters around standard screen sizes, and potential outliers.

c. Draw a sketch of the chart



d. Draw a sketch of the data table required to build the chart

Rows: 4740 | Columns: 2

Sorting

Filter

Search

⌵

×

☐	#	RowID	Avg_mode_power Number (Float)	screensize_inches Number (Float)
☐	4236	Row4...	13.7	19
☐	1711	Row7...	14	24
☐	2756	Row2...	14.7	16
☐	2008	Row4...	14.76	19
☐	1704	Row3...	14.89	19
☐	1268	Row1...	15.1	24
☐	1273	Row1...	15.1	24
☐	1274	Row1...	15.1	24
☐	1292	Row9...	15.1	24
☐	1297	Row1...	15.1	24
☐	2754	Row1...	15.1	16
☐	2755	Row2...	15.1	16
☐	2906	Row2...	15.1	24

Answer: The larger the screen size, the higher the power use.

6. What is the relationship between star rating and screen size?

a. What data is needed to answer this question?

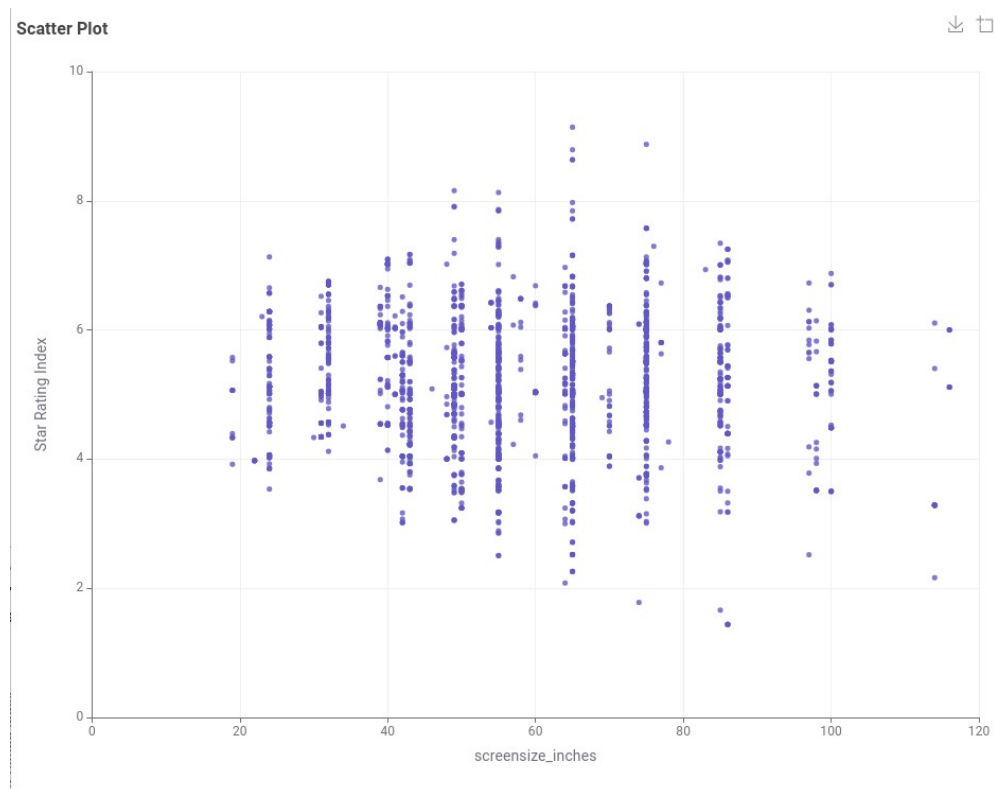
Column Name	Data Type	Description
Star Rating Index	Float	A continuous numerical score indicating the energy efficiency rating or performance index of the screen.
screensizes -> inches	Float	The physical diagonal display dimension of each screen measured in inches.

b. What chart type would be good to represent this data and give some reasons?

Scatter Plot Chart

Reason: A scatter plot is well suited for examining the potential relationship or correlation between two continuous variables (screen size and star rating index). It plots each screen as an individual coordinate, clearly demonstrating the distribution, density, and whether energy efficiency ratings systematically vary with physical display size.

c. Draw a sketch of the chart



d. Draw a sketch of the data table required to build the chart

Rows: 4740 Columns: 2 Sorting Filter Search ⌵ ×				
☐	#	RowID	Star Rating Index Number (Float)	screensize_inches Number (Float)
☐	3797	Row3...	1.01	65
☐	3798	Row3...	1.01	65
☐	3799	Row3...	1.01	65
☐	3788	Row2...	1.069	65
☐	3789	Row2...	1.069	65
☐	3790	Row2...	1.069	65
☐	3619	Row3...	1.093	55
☐	3971	Row3...	1.156	75
☐	3972	Row3...	1.156	75
☐	3973	Row3...	1.156	75
☐	3809	Row3...	1.436	65
☐	3810	Row3...	1.436	65
☐	3811	Row3...	1.436	65
☐	1121	Row3...	1.441	86

Answer: Based on the data analysis, it doesn't show any relationship or linear correlation between screen size and star rating. A larger TV screen does not systematically result in a higher or lower Star Rating index.

[optional] Are there differences in power consumption between brands?

a. What data is needed to answer this question?

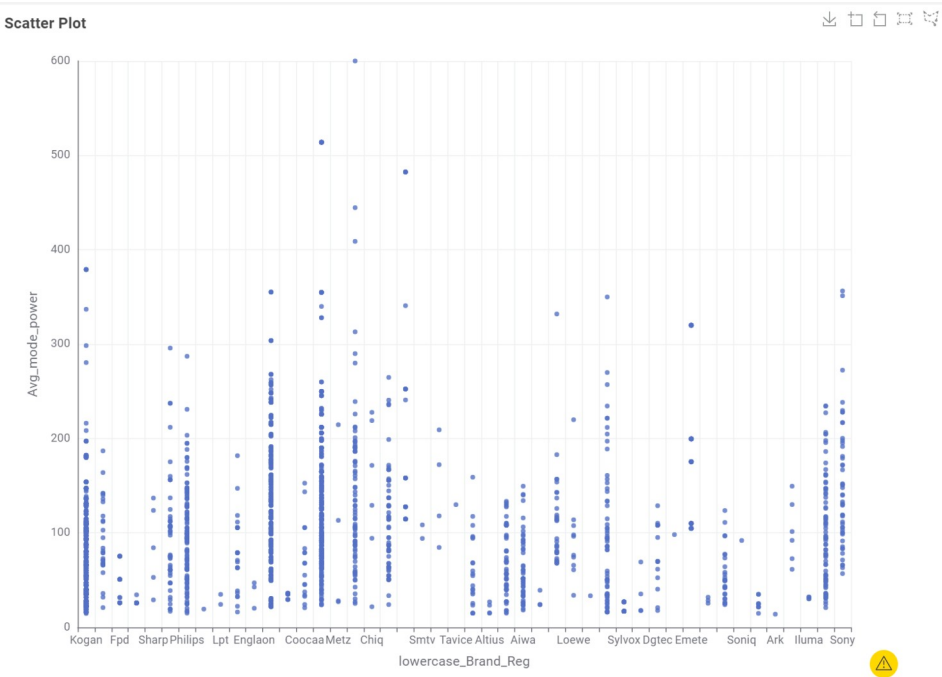
Column Name	Data Type	Description
Brand_Reg (Lowered case)	String	The standardized brand name recorded in lowercase text to ensure consistent grouping and accurate count aggregation.
Avg_mode_power	Float	The continuous numerical value representing average power consumption in on-mode.

b. What chart type would be good to represent this data and give some reasons?

Scatter Plot Chart

Reason: A scatter plot (strip plot) effectively displays the full distribution of a continuous numerical variable (Avg_mode_power) across discrete categorical groups (Brand_Reg). By plotting every individual model as a distinct point along each brand's column, it highlights the range, spread, density clusters, and extreme high-power outliers within each brand rather than hiding variance behind a single average.

c. Draw a sketch of the chart



d. Draw a sketch of the data table required to build the chart

Rows: 4740 | Columns: 2 | Sorting | Filter

Search

	#	RowID	lowercase_Brand_Reg String	Avg_mode_power Number (Float)
☐	1	Row2...	Kogan	25.3
☐	2	Row2...	Kogan	35.2
☐	3	Row1...	Kogan	110.3
☐	4	Row3...	Samsung	36
☐	5	Row4...	Fpd	26.06
☐	6	Row4...	Fpd	31.68
☐	7	Row4...	Fpd	50.94
☐	8	Row4...	Fpd	75.42
☐	9	Row1...	Caixun	25.92
☐	10	Row1...	Kogan	34.52
☐	11	Row4...	Kogan	36.3
☐	12	Row4...	Kogan	45.18
☐	13	Row4...	Kogan	62.52
☐	14	Row4...	Kogan	65.04
☐	15	Row4...	Kogan	80.58

Answer: The differences show no relationship between power consumption and brands.

T03: Communicating Data Insights

- **Link to your [Miro board](#)****Links to an external site.** showing the storyboard for answering the 6 questions